InfraWatch

InfraWatch is a centralized server management platform designed to help administrators monitor and manage multiple Linux and Windows servers from a single web interface. The system provides real-time health monitoring, remote configuration capabilities, automated alerts, and centralized logging. InfraWatch simplifies the complex task of managing distributed server infrastructure by offering an intuitive dashboard where you can view system metrics, execute commands, manage network settings, and receive notifications about potential issues before they become critical problems.

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Installation - Docker Method (Recommended)

The Docker-based installation is the easiest way to get InfraWatch running. It automatically sets up all required components including the database, backend server, and frontend interface.

Prerequisites

Before starting, you need to install the following software on your computer:

For Windows Users:

6. **Git for Windows** - Download from https://git-scm.com/download/win
7. **Docker Desktop** - Download from https://www.docker.com/products/docker-desktop
8. **Windows Subsystem for Linux (WSL2)** - Open PowerShell as Administrator and run:

```
wsl --install
```

9. **Enable Virtualization** - This must be enabled in your computer's BIOS settings. Restart your computer, press the BIOS key (usually F2, F10, or DEL during startup), and enable "Intel VT-x" or "AMD-V" depending on your processor.

For Linux Users:

10. **Git** - Install using your package manager:

```
# For Ubuntu/Debian
sudo apt update
sudo apt install git

# For Fedora/RHEL
sudo dnf install git

# For Arch Linux
sudo pacman -S git
```

11. **Docker and Docker Compose** - Install using:

```
# For Ubuntu/Debian

sudo apt update
sudo apt install docker.io docker-compose
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker $USER#




For Fedora/RHEL
sudo dnf install docker docker-compose
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker $USER#




For Arch Linux
sudo pacman -S docker docker-compose
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker $USER
```

Important: After running these commands, log out and log back in for the group changes to take effect.

Step 1: Download the Project

You need to download the InfraWatch project files to your computer. There are two methods:

Method A: Using Git (Recommended)

Open a terminal (Linux) or Command Prompt/PowerShell (Windows) and run:

```
git clone https://github.com/K1sh0rx/InfraWatch.git
```

This command downloads the entire project into a folder called "InfraWatch" in your current directory.

Method B: Download as ZIP

12. Go to https://github.com/K1sh0rx/InfraWatch
13. Click the green "Code" button
14. Select "Download ZIP"
15. Extract the ZIP file to a location of your choice

Step 2: Navigate to the Project Directory

After downloading, you need to open a terminal inside the project folder.

For Linux:

```
cd InfraWatch
```

For Windows (PowerShell):

```
cd InfraWatch
```

For Windows (Command Prompt):

```
cd InfraWatch
```

To verify you are in the correct directory, list the files:

For Linux:

```
ls
```

For Windows (PowerShell):

```
dir
```

You should see output similar to this:

```
Backend

/client
```

```
/db
/docker-compose.yml
/docs
/frontend
/README.md
setup.ps1
setup.sh
```

If you see these folders and files, you are in the correct location.

Step 3: Run the Setup Script

The setup script will guide you through the installation process. It will ask you a few questions and then automatically configure everything.

For Linux:

First, make the script executable:

```
chmod +x setup.sh
```

Then run it:

```
./setup.sh start
```

For Windows (PowerShell):

Run PowerShell as Administrator (right-click PowerShell and select "Run as Administrator"), then:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
.\setup.ps1 start
```

Step 4: Answer the Setup Questions

The script will ask you several questions. Here's what each means and how to answer:

Question 1: Select Host IP Address

The script will show you a list of IP addresses available on your computer:

```
Available host IP addresses:
1) 192.168.1.100   (default/active)
2) 10.0.0.50
Select host IP [1-2] (default: 1):
```

What this means: The IP address is how other computers on your network will access InfraWatch. If you're just testing on your own computer, you can press Enter to accept the default. If you want other computers to access it, choose the IP address that's on your local network (usually starts with 192.168).

How to answer: Press Enter to use the default, or type the number corresponding to your preferred IP and press Enter.

Question 2: Enter DATABASE Port

```
Enter DATABASE port (default: 9001):
```

What this means: This is the network port where the PostgreSQL database will listen for connections. The database stores all your server information, metrics, and logs.

How to answer: Press Enter to use the default port 9001, or type a different port number if 9001 is already in use by another program on your computer.

Question 3: Enter BACKEND Port

```
Enter BACKEND port (default: 9000):
```

What this means: This is the port where the InfraWatch backend API server will run. The frontend communicates with the backend through this port.

How to answer: Press Enter to use the default port 9000, or type a different port number if needed.

Step 5: Wait for Installation to Complete

After answering the questions, the script will:

16. Generate configuration files (.env files) with your settings
17. Download and build Docker images (this may take 5-10 minutes the first time)
18. Start the database, backend, and frontend containers
19. Initialize the database with the required tables and structure

You will see various messages scrolling on the screen. This is normal. Wait until you see:

```
System setup complete!
Frontend:    http://192.168.1.100
Backend API: http://192.168.1.100:9000
PostgreSQL:  Port 9001
```

The IP addresses and ports shown will match what you selected during setup.

Step 6: Access InfraWatch

Open your web browser and go to the Frontend URL shown in the completion message. For example:

```
http://192.168.1.100
```

You should see the InfraWatch login page.

Default Login Credentials:

- Username: `admin`
- Password: `admin`

Important: Change the default password immediately after your first login for security reasons.

Installation - Manual Method

The manual installation method gives you more control over the setup process but requires more technical knowledge and manual configuration. Use this method if you need to customize the installation or if you prefer not to use Docker.

Prerequisites

Before starting manual installation, ensure you have the following software installed:

20. **PostgreSQL Database (version 15 or higher)**
    - Download from https://www.postgresql.org/download/
21. **Go Programming Language (version 1.20 or higher)**
    - Download from https://go.dev/dl/
22. **Node.js JavaScript Runtime (version 18 or higher)**
    - Download from https://nodejs.org/
23. **sqlc Code Generator**
    - Install using:

```
go install github.com/sqlc-dev/sqlc/cmd/sqlc@latest
```

24. **Git Version Control**
    - Download from https://git-scm.com/

Step 1: Download the Project

Clone the repository:

```
git clone https://github.com/K1sh0rx/InfraWatch.git
cd InfraWatch
```

Step 2: Set Up PostgreSQL Database

Start the PostgreSQL service:

For Linux:

```
sudo systemctl start postgresql
sudo systemctl enable postgresql
```

For Windows:

Create a database user and database:

```
# Open PostgreSQL command line# For Linux:
sudo -u postgres psql
# For Windows:#
Search for "SQL Shell (psql)" in the Start menu and open it# Press Enter for
default values, then enter the password you set during PostgreSQL installation
```

Once in the PostgreSQL prompt, run these commands:

```
CREATE USER admin WITH PASSWORD 'admin';
CREATE DATABASE smsdb;
GRANT ALL PRIVILEGES ON DATABASE smsdb TO admin;
\c smsdb
GRANT ALL ON SCHEMA public TO admin;
```

To exit PostgreSQL:

```
\q
```

Test the connection:

```
psql -U admin -d smsdb -h localhost -p 5432
```

When prompted, enter the password: `admin`If you successfully connect, type `\q` to exit.

Step 3: Configure the Backend

Navigate to the backend directory:

```
cd backend
```

Generate database access code using sqlc:

```
cd dbsqlc generatecd ..
```

Create a file named `.env` in the `backend` directory with the following content:

```
DATABASE_URL=postgres://admin:admin@localhost:5432/smsdb?sslmode=disable
CLIENT_PORT=2210
CLIENT_PROTOCOL=http
JWT_SECRET=xxxxxxxxxxxxxxxx
SERVER_PORT=8000
LOG_LEVEL=info
```

```
SMTP_HOST=smtp.google.com
SMTP_PORT=574
SMTP_USERNAME=xxxxxxxxxxxxx@gmail.com
SMTP_PASSWORD=xxxxxxxxxxxxxx
SMTP_FROM=SMS Alerts <xxxxxxxxxxxxx@gmail.com>
```

Note: Replace the database connection details if you used different values during PostgreSQL setup.

Initialize the database structure:

```
go run temp/dbinit.go
```

You should see messages indicating that tables were created successfully.

Build the backend server:

```
go build -o server main.go
```

This creates an executable file named `server` (or `server.exe` on Windows).

Step 4: Configure the Frontend

Open a new terminal window and navigate to the frontend directory:

```
cd frontend
```

Install the required Node.js packages:

```
npm install
```

This will download all dependencies. It may take a few minutes.

Create a file named `.env` in the `frontend` directory:

```
VITE_BACKEND_URL=http://localhost:9000
```

Note: Replace `localhost` with your computer's IP address if you want to access InfraWatch from other computers on your network.

Build the frontend:

```
npm run build
```

Step 5: Start the Services

You need to start both the backend and frontend in separate terminal windows.

Terminal 1 - Start Backend:

```
cd backend./server
```

On Windows:

```
cd backendserver.exe
```

You should see output like:

```
Server running on port 8000
Database connected successfully
```

Terminal 2 - Start Frontend:

```
cd frontend
npm run preview
```

You should see output like:

```
Local:   http://localhost:4173/
Network: http://192.168.1.100:4173/
```

Step 6: Access InfraWatch

Open your web browser and navigate to the URL shown in the frontend terminal (usually http://localhost:4173).

Default Login Credentials:

- Username: `admin`
- Password: `admin`

Important: Change the default password immediately after your first login.

Server Controller Setup

The Server Controller is a lightweight program that runs on each server you want to manage with InfraWatch. It communicates with the central backend to send health metrics and receive commands.

Prerequisites

The server you want to monitor must have:

- Go programming language (version 1.20 or higher)
- Git
- Network connectivity to the InfraWatch backend server

Step 1: Download the Project on the Target Server

On the server you want to monitor, run:

```
git clone https://github.com/K1sh0rx/InfraWatch.git
cd InfraWatch
```

Step 2: Navigate to the Correct Client Directory

For Linux Servers:

```
cd client/linux
```

For Windows Servers:

```
cd client\windows
```

Step 3: Install Dependencies

```
go mod tidy
```

This command downloads all required Go packages for the controller.

Step 4: Build the Controller

For Linux:

```
go build -o controller main.go
```

For Windows:

```
go build -o controller.exe main.go
```

Step 5: Generate an Access Token

Before running the controller, you need to generate an access token from the InfraWatch admin panel:

25. Log in to the InfraWatch web interface
26. Navigate to "Configuration" or "Settings"
27. Click on "Server Management" or "Add New Server"
28. Click "Generate Token" or "Create Access Token"
29. Copy the generated token (it will look like a long random string)

Step 6: Run the Controller

For Linux:

```
./controller
```

For Windows:

```
controller.exe
```

On first run, the controller will prompt you to enter the access token:

```
Enter access token:
```

Paste the token you copied from the admin panel and press Enter.

The controller will save this token and use it for all future connections. You should see output like:

```
Controller started successfully
Connected to backend: http://192.168.1.100:9000
Sending health metrics every 30 seconds
```

Step 7: Verify Connection

Return to the InfraWatch web interface and check the server list. You should see your newly connected server appear with its hostname, IP address, and current status.

First Time Access

After successful installation, follow these steps to get started with InfraWatch:

Step 1: Initial Login

30. Open your web browser
31. Navigate to the InfraWatch URL (e.g., http://192.168.1.100)
32. You will see the login page
33. Enter the default credentials:
    - Username: `admin`
    - Password: `admin`
34. Click "Login"

Step 2: Change Default Password

For security reasons, you should immediately change the default password:

35. After logging in, look for "Settings" in the navigation menu
36. Click on "User Management" or "Account Settings"
37. Find the "Change Password" option
38. Enter a strong new password
39. Confirm the new password
40. Click "Save" or "Update Password"

Step 3: Add Your First Server

41. Navigate to "Configuration" or "Servers" in the menu
42. Click "Add New Server" or "Register Device"
43. Fill in the server details:
    - **Server Name:** A friendly name for identification (e.g., "Web Server 1")
    - **IP Address:** The IP address of the server you want to monitor
    - **Operating System:** Select Linux or Windows
44. Click "Generate Access Token"
45. Copy the generated token

46. Follow the Server Controller Setup instructions to install the controller on that server
47. When prompted, paste the access token

Step 4: Explore the Dashboard

After adding a server, explore the different sections:

- **Health Dashboard:** View real-time CPU, memory, disk, and network metrics
- **Alerts:** See any warnings or critical issues detected on your servers
- **Logs:** Browse system logs from all connected servers
- **Configuration:** Manage server settings, network configuration, and firewall rules
- **Resource Optimization:** View and manage running services, perform cleanup operations

Troubleshooting

Docker Installation Issues

Problem: "docker: command not found"

Solution: Docker is not installed or not in your system PATH.

- Verify Docker installation: `docker --version`
- Reinstall Docker if necessary
- On Linux, ensure Docker service is running: `sudo systemctl start docker`

Problem: "permission denied while trying to connect to the Docker daemon"

Solution: Your user does not have permission to use Docker.

- On Linux: `sudo usermod -aG docker $USER`
- Log out and log back in
- Verify with: `groups` (you should see "docker" in the list)

Problem: Setup script fails with "port already in use"

Solution: Another program is using the required port.

- Check what's using the port: `sudo lsof -i :9000` (replace 9000 with your port)
- Either stop that program or run the setup script again and choose a different port

Problem: Database initialization failed

Solution: The database container may not be ready yet.

- Wait 30 seconds and run: `docker exec sms-backend ./dbinit`
- If it still fails, check logs: `docker logs sms-backend`

Manual Installation Issues

Problem: "psql: error: connection to server failed"

Solution: PostgreSQL is not running or connection details are incorrect.

- Check if PostgreSQL is running:
  - Linux: `sudo systemctl status postgresql`
  - Windows: Check Services panel
- Verify port 5432 is open: `netstat -an | grep 5432`
- Check connection string in backend/.env matches your PostgreSQL setup

Problem: "sqlc: command not found"

Solution: sqlc is not installed or not in PATH.

- Install sqlc: `go install github.com/sqlc-dev/sqlc/cmd/sqlc@latest`
- Ensure Go bin directory is in PATH: `export PATH=$PATH:$(go env GOPATH)/bin`
- Verify installation: `sqlc version`

Controller Issues

Problem: "Connection refused" when starting controller

Solution: Backend server is not accessible.

- Verify backend is running and accessible
- Check if the IP address and port are correct
- Test connection: `curl http://YOUR_BACKEND_IP:9000/health` (Linux/Mac) or use browser
- Check firewall rules on both controller and backend machines

Getting Help

If you encounter issues not covered here:

48. Check the project repository issues: https://github.com/K1sh0rx/InfraWatch/issues
49. Search for similar problems others may have reported
50. Create a new issue with:
    - Detailed description of the problem
    - Steps to reproduce
    - Error messages (if any)
    - Your operating system and versions
    - Installation method used (Docker or Manual)

Support and Documentation

For additional help and documentation:

- Project Repository: https://github.com/K1sh0rx/InfraWatch
- Issues and Bug Reports: https://github.com/K1sh0rx/InfraWatch/issues
- Full Documentation: See the `docs` folder in the project directory

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